1.1 The Problem

College students have to manage their daily routines without relying on their parents to wake them up or keep them on schedule. Students may sleep through alarms or repeatedly snooze them, causing them to miss important classes or exams.

1.2 The Solution

I am going to build a shared alarm app that helps college students wake up for classes and exams. If a student does not confirm they are awake, the app will notify a chosen friend who can check on them.

2.1 Eisenhower Matrix

	Urgent	**Not Urgent**
Important	**Do – Your MVP** ● Let users set an alarm with a time and a label, such as "Math Exam." ● Give users a way to confirm they are awake when the alarm goes off. ● Let users enter an accountability partner's phone number. ● Notify that partner if the user does not confirm they are awake within a set time.	**Schedule – For V2.0+** ● Add recurring alarms for weekly class schedules. ● Let users choose different accountability partners for different alarms. ● Track missed alarms and successful wake-ups. ● Allow partners to confirm they have checked on the user. ● Can be used to keep track of regular tasks as well.
Not Important	**Delegate – Use a Library** ● Use a scheduling library to trigger alarms at the correct	**Delete – Cut It** ● Add a public leaderboard ranking students by

	time. ● Use an audio library to play alarm sounds. ● Use an email library or service to send notifications to accountability partners.	wake-up time. ● Include a social feed for posting daily routines. ● Add an alarm sound marketplace.

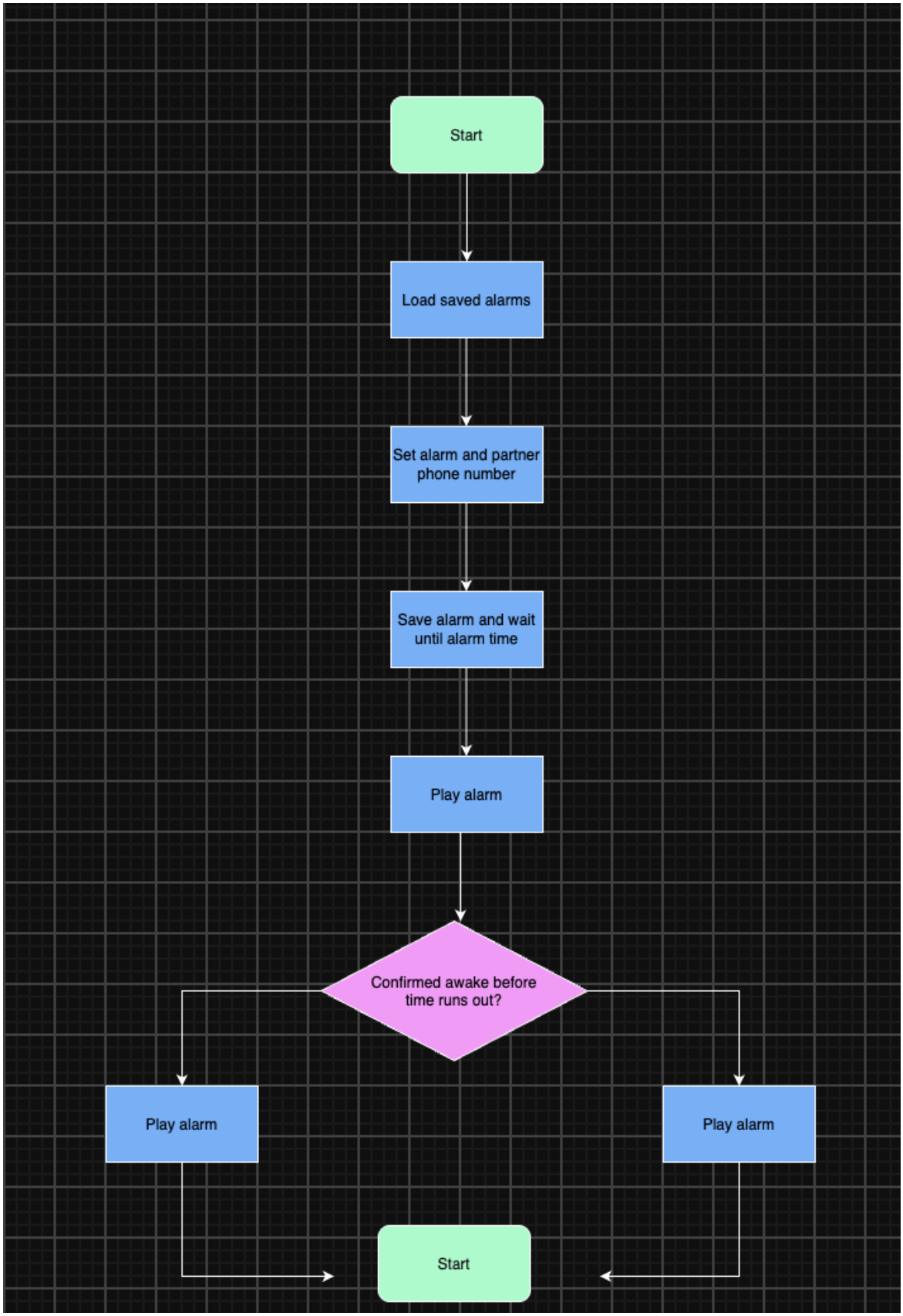
2.2 Tech Stack

- Languages: Java for the program's logic, HTML for an interface mockup, and CSS for styling.
- Tools and Java features: Git and GitHub for tracking changes and storing the project; arrays, ArrayLists, loops, and classes for organizing and managing alarms.
- Data storage: A plain text file to save alarm times, labels, and accountability partners' phone numbers.
- External services: No outside datasets are needed. Sending actual text notifications will require learning and testing an SMS service.

2.3 Chunked Build Plan

1. Add starter project files and a README to the GitHub repository.
2. Create an Alarm class and an ArrayList to store alarms.
3. Add a terminal menu for entering and viewing alarm times, labels, and partners' phone numbers.
4. Add saving and loading alarms using a text file.
5. Make alarms trigger at the scheduled time and play a sound.
6. Add an "I'm awake" confirmation and a response time limit.
7. Connect and test an SMS service to notify the partner if the user does not respond.
8. Create an HTML and CSS mockup of the interface.
9. Test the full process, fix remaining bugs, and add running instructions to the README.

2.4 System Diagram


Start
Load saved alarms
Set alarm and partner phone number
Save alarm and wait until alarm time
Play alarm
Confirmed awake before time runs out?
Play alarm
Play alarm
Start

3.1 GitHub Repository
https://github.com/ruhirdesai/accountability_project